Sen, *Collective Choice and Social Welfare*: Ch 1*, 2*, A1*

Definitions, examples, and results

A reference sheet for the starred chapters. Throughout, X is the set of alternatives (social states), $S \subseteq X$, and R is a weak preference relation ("at least as good as") on X. Individual i's relation is R_i.

Part 1: Definitions

1.1 Properties of a binary relation (Ch 1*, §1*1)

For a binary relation R on a set S:

Table 1: The six properties

Property	Statement
Reflexivity	$\forall x \in S : xRx$
Completeness	$\forall x, y \in S : (x \neq y) \rightarrow (xRy \vee yRx)$
Transitivity	$\forall x, y, z \in S : (xRy \wedge yRz) \rightarrow xRz$
Anti-symmetry	$\forall x, y \in S : (xRy \wedge yRx) \rightarrow x = y$
Asymmetry	$\forall x, y \in S : xRy \rightarrow \sim (yRx)$
Symmetry	$\forall x, y \in S : xRy \rightarrow yRx$

Asymmetry implies anti-symmetry, not conversely. (If $xRy \rightarrow \sim (yRx)$ then the antecedent of anti-symmetry is never satisfied, so it holds vacuously.)

1.2 Named types of relation (Ch 1*, §1*1)

Table 2: Sen's terminology

#	Properties satisfied	Sen's name	Other names in the literature
1	reflexivity, transitivity	**quasi-ordering**	pre-ordering
2	reflexivity, transitivity, completeness	**ordering**	complete pre-ordering; complete quasi-ordering; weak ordering
3	reflexivity, transitivity, anti-symmetry	**partial ordering**	ordering
4	reflexivity, transitivity, completeness, anti-symmetry	**chain**	linear ordering; complete ordering; simple ordering
5	transitivity, asymmetry	**strict partial ordering**	—
6	transitivity, asymmetry, completeness	**strong ordering**	ordering; strict ordering; strict complete ordering

The terminology for these is all over the place, but these are Sen's terms, and we'll use them.

1.3 Strict preference, indifference, maximality, choice (Ch 1*, §1*2)

Definition 1. ***Definition 1*1.*** $xPy \leftrightarrow [xRy \wedge \sim (yRx)]$

Definition 2. ***Definition 1*2.*** $xIy \leftrightarrow [xRy \wedge yRx]$

Definition 3. ***Definition 1*3.*** *$x \in S$ is a **maximal element** of S with respect to R iff* $\sim [\exists y : (y \in S \wedge yPx)]$.

*The set of maximal elements of S is the **maximal set**, $M(S, R)$.*

Definition 4. ***Definition 1*4.*** *$x \in S$ is a **best element** of S with respect to R iff* $\forall y : (y \in S \rightarrow xRy)$.

*The set of best elements of S is the **choice set**, $C(S, R)$.*

1.4 Subrelations and compatibility (Ch 1*, §1*4)

Definition 5. ***Definition 1*5.*** *A quasi-ordering Q_1 is a **subrelation** of Q_2 iff for all $x, y \in X$:*

1. $xQ_1y \rightarrow xQ_2y$
2. $[xQ_1y \wedge \sim (yQ_1x)] \rightarrow \sim (yQ_2x)$

That is: whatever Q_1 says, Q_2 says too, both about weak preference and about strict preference.

Definition 6. ***Definition 1*6.*** *If a quasi-ordering Q is a subrelation of an ordering R, then R is* ***compatible*** *with Q.*

Definition 7. ***Definition 1*7.*** *Two quasi-orderings Q_1 and Q_2 are* ***compatible*** *iff there is an ordering compatible with each.*

1.5 Choice functions and weakened transitivity (Ch 1*, §1*5)

Definition 8. ***Definition 1*8.*** *A* ***choice function*** *$C(S, R)$ defined over X is a functional relation such that $C(S, R)$ is non-empty for every non-empty $S \subseteq X$.*

Definition 9. ***Definition 1*9.*** *R is* ***quasi-transitive*** *iff for all $x, y, z \in X$: $(xPy \wedge yPz) \rightarrow xPz$.*

Definition 10. ***Definition 1*10.*** *R is* ***acyclical*** *over X iff*

$$\forall x_1, ..., x_j \in X : [(x_1 P x_2 \wedge x_2 P x_3 \wedge ... \wedge x_{j-1} P x_j) \rightarrow x_1 R x_j]$$

1.6 Rationality conditions on choice functions (Ch 1*, §1*6)

Here $C(S)$ is any choice function over X, not necessarily derived from a binary relation.

Property α: $x \in S_1 \subseteq S_2 \rightarrow [x \in C(S_2) \rightarrow x \in C(S_1)]$

Property β: $[x, y \in C(S_1) \wedge S_1 \subseteq S_2] \rightarrow [x \in C(S_2) \leftrightarrow y \in C(S_2)]$

Definition 11. ***Definition 1*11.*** *R is* ***PI-transitive*** *over X iff for all $x, y, z \in X$: $(xPy \wedge yIz) \rightarrow xPz$.*

The four mixed transitivity conditions named in Lemma 1*a, used throughout:

Table 3: Mixed transitivity conditions

Label	Condition
II	$(xIy \wedge yIz) \rightarrow xIz$
PI	$(xPy \wedge yIz) \rightarrow xPz$
IP	$(xIy \wedge yPz) \rightarrow xPz$
PP	$(xPy \wedge yPz) \rightarrow xPz$ (i.e. quasi-transitivity)

Part 1 (continued): Chapter 2* definitions

1.7 Collective choice rules (Ch 2*, §2*1)

X is the set of social states, R_i is person i's preference relation, $i = 1, ..., n$. Each R_i is assumed to be an ordering. R is social preference, and **no such assumption is made about R**. Whether R is an ordering is the question under investigation.

Definition 12. ***Definition 2*1.*** *A* ***collective choice rule*** *(CCR) is a functional relation f such that for any set of n individual orderings $R_1, ..., R_n$, one and only one social preference relation R is determined, $R = f(R_1, ..., R_n)$.*

Definition 13. ***Definition 2*2.*** *A collective choice rule is* ***decisive*** *iff its range is restricted to complete preference relations R.*

1.8 The Pareto relations (Ch 2*, §2*1)

Definition 14. ***Definition 2*3.*** *For all $x, y \in X$:*

1. $x\bar{R}y \leftrightarrow [\forall i : xR_iy]$
2. $x\bar{P}y \leftrightarrow [x\bar{R}y \wedge \sim (y\bar{R}x)]$
3. $x\bar{I}y \leftrightarrow [x\bar{R}y \wedge y\bar{R}x]$

So $x\bar{P}y$ holds when everyone weakly prefers x and at least one person strictly prefers it.

Definition 15. ***Definition 2*4.*** *For all $x, y \in X$: $x\bar{\bar{P}}y \leftrightarrow \forall i : xP_iy$*

Definition 16. ***Definition 2*5.*** *A collective choice rule is* ***Pareto-inclusive*** *iff its range is restricted to social preference relations R such that $\bar{R}$ is a subrelation of R, i.e.*

$$\forall x, y \in X : [(x\bar{R}y \rightarrow xRy) \wedge (x\bar{P}y \rightarrow xPy)]$$

Definition 17. ***Definition 2*6.*** *For any n-tuple $(R_1, ..., R_n)$, a state $x \in S$ is* ***Pareto-optimal*** *in S iff $\sim [\exists y \in S : y\bar{P}x]$. A Pareto-optimal state is also called* ***economically efficient****.*

Part 1 (continued): Chapter A1* definitions

1.10 Arrow's framework (Ch A1*)

A **social welfare function** (SWF), in Arrow's sense, is a functional relation specifying one social ordering R for any n-tuple of individual orderings $\{R_i\}$: $R = f(\{R_i\})$. (Contrast Definition 2*1: a CCR need not deliver an *ordering*.)

Arrow's choice set, defined purely relationally:

$$C(S) = [x \mid x \in S \wedge \forall y \in S : xRy]$$

Condition I^A (Arrow's independence of irrelevant alternatives). For any two n-tuples $\{R_i\}$ and $\{R'_i\}$ in the domain of f, and for any $S \subseteq X$, with choice functions $C(\cdot)$ and $C'(\cdot)$ corresponding to $\{R_i\}$ and $\{R'_i\}$:

$$[\forall i : (\forall x, y \in S : xR_iy \leftrightarrow xR'_iy)] \Rightarrow C(S) = C'(S)$$

Condition I (relational independence). The restriction of the social preference relation over any pair $\{x, y\}$ is a function of the n-tuple of restrictions of individual preferences over that pair:

$$R^{\{x,y\}} = f^{\{x,y\}}(\left\{R_i^{\{x,y\}}\right\})$$

Given Arrow's relational definition of $C(S)$, I and I^A are equivalent. Sen uses I, but it isn't equivalent to I^A without a different definition of $C(S)$.

Arrow's axioms

- **U (Unrestricted Domain).** For any logically possible set of individual preferences, there is a social ordering R.
- $\boldsymbol{I^2}$ **(Independence of Irrelevant Alternatives).** The social ranking of any pair $\{x, y\}$ depends only on the individual rankings of x and y.
- **P (Pareto Principle).** If everyone prefers x to y, then x is socially preferred to y.
- **D (Non-dictatorship).** There is no person i such that whenever i prefers x to y, x is socially preferred to y, no matter what others prefer.

Background assumptions: at least three distinct social states, and a finite number (≥ 2) of individuals.

Decisiveness

A set $G \subseteq H$ of individuals (where H is the set of all individuals) is **locally decisive over $\{x, y\}$** iff x is socially preferred to y whenever all individuals in G prefer x to y, no matter what the others prefer. G is **(globally) decisive** iff it is locally decisive over every pair.

A person i is **semi-decisive** over an ordered pair $\{x, y\}$ iff xP_iy implies xRy. A person **has a veto** iff he or she is semi-decisive over every ordered pair.

Weakened consistency conditions

Quasi-transitivity: $(xPy \wedge yPz) \rightarrow xPz$. Indifference need not be transitive, and mixed cases need not yield strict preference.

Suzumura consistency: for any sequence of weak preferences $x_1Rx_2, x_2Rx_3, ..., x_{j-1}Rx_j$, we must not have x_jPx_1.

Semi-transitivity: $\forall x, y, z, a \in X$: if xPy and yPz, then xPa or aPz.

Interval order property: $\forall x, y, a, b \in X$: if xPy and aPb, then xPb or aPy.

A **semi-order** satisfies both (in addition to quasi-transitivity, which each of them implies for complete R).

An $\boldsymbol{s}$**-and-**$\boldsymbol{t}$ **order** generalises: $P^sIP^t \Rightarrow P$, with s strict preferences preceding the indifference and t following it. Orderings are the case $s + t = 1$; semi-orders are $s + t = 2$.

Rules and structures

A **social decision function** (SDF) is obtained from an Arrovian SWF by dropping the requirement that R be an ordering, demanding only that R be reflexive and complete and generate a choice function — i.e. that $C(S, R)$ be non-empty for every non-empty S.

The **Pareto-extension rule**: x is socially preferred to y iff everyone prefers x to y; and x and y are socially indifferent if they are either Pareto-indifferent *or* Pareto-incomparable.

An SDF is **oligarchic** (Gibbard 1969) iff there is a unique group G such that G as a whole group is fully decisive, and every member of G has a veto.

Condition NIM (neutrality, independence with monotonicity) (Blau and Deb 1977). For any $x, y, a, b \in X$ and any two profiles $\{R_i\}$, $\{R'_i\}$: if for all i, $xP_iy \Rightarrow aP'_ib$ and $xI_iy \Rightarrow aR'_ib$, then $xPy \Rightarrow aP'b$.

The topology of decisive sets

Let Ω be the class of decisive sets of individuals — a subset of the power set of H. Consider:

1. $H \in \Omega$
2. $[G \in \Omega \wedge G \subseteq J] \Rightarrow J \in \Omega$
3. $[G_1, ..., G_k \in \Omega$ for finite $k] \Rightarrow$ the intersection of these G_i is non-empty
4. $[G, J \in \Omega] \Rightarrow G \cap J \in \Omega$
5. $[G \in \Omega] \Rightarrow H - G \notin \Omega$

Ω is a **pre-filter** iff it satisfies (1)–(3); a **filter** iff it also satisfies (4); an **ultra-filter** iff it satisfies all of (1)–(5).

A **collegium** (Brown) is a group of persons such that every member of it belongs to every decisive set of persons. Where Ω is a pre-filter, (3) guarantees a collegium exists; where Ω is a filter, (4) makes the collegium itself decisive, defining the oligarchy.

Part 2: Real-world examples of the relation types

2.1 Sen's own examples

"Is at least as tall as", on mountain peaks with measured heights. Reflexive (a peak is as tall as itself); complete (if A is not at least as tall as B, then B is taller); transitive; *not* anti-symmetric (two distinct peaks can be the same height); not asymmetric; not symmetric. So: an **ordering**, but not a chain and not a partial ordering.

"Is taller than", on the same set. Transitive, anti-symmetric, asymmetric; not reflexive, not complete, not symmetric. So: a **strict partial ordering**, but not a strong ordering, because ties make it incomplete.

"Is brother of", on men (Sen's footnote 1). Commonly *thought* transitive, but is not: A brother of B and B brother of A would give, by transitivity, A brother of himself.

2.2 Chains (reflexive, transitive, complete, anti-symmetric)

"Is at least as far forward in the queue as", over the people currently in a single queue. Positions are integers and no two people occupy the same one, so anti-symmetry holds.

"Is alphabetically no later than", over a set of distinct strings. (It's because it's defined on *strings* and not *words* that we can't have ties.)

2.3 Orderings (reflexive, transitive, complete; ties allowed)

"Scored at least as many points as", over the teams in a league table at a given moment.

"Weighs at least as much as", over a set of objects.

2.4 Partial orderings (reflexive, transitive, anti-symmetric, incomplete)

Set inclusion $\subseteq$, over the subsets of any set with more than one element. One obvious instance of incompleteness: $\{a\} \not\subseteq \{b\}$ and $\{b\} \not\subseteq \{a\}$, so it is incomplete; and $A \subseteq B$ with $B \subseteq A$ gives $A = B$, so it is anti-symmetric.

"Is an ancestor of, or identical to", over the members of a family tree. Reflexive by construction, transitive, anti-symmetric, and incomplete (siblings are incomparable).

"Divides", over the positive integers. Reflexive, transitive, anti-symmetric, incomplete (6 and 10 are incomparable).

Course prerequisite structure, over the courses in a department, taking the transitive-reflexive closure. Anti-symmetric if and only if the prerequisite graph has no cycles, which would be hilarious if it was violated.

2.5 Strict partial orderings (transitive, asymmetric)

"Is a proper subset of", over subsets. The strict counterpart of §2.4.

"Is an ancestor of", over a family tree.

The Pareto relation $\bar{P}$ (Lemma 2*c).

"Is strictly north of", over points on a map. Transitive, asymmetric, incomplete (two points at the same latitude).

2.6 Quasi-orderings (reflexive, transitive, incomplete)

This is the type Sen cares most about, and the one he thinks social choice theory has been too quick to strengthen.

The Pareto relation $\bar{R}$ (Lemma 2*a). Reflexive and transitive for every logically possible profile; complete only under the very strong condition of Lemma 2*b (that whenever anyone strictly prefers x to y, everyone weakly prefers x to y). Note that $\bar{R}$ is a quasi-ordering but *not* a partial ordering: two distinct Pareto-indifferent states are not identical, so anti-symmetry fails.

Logical strength: "entails", over propositions or theories. Reflexive, transitive, incomplete.

"Is at least as good on every criterion as", over any multi-criterion comparison.

"Is at least as well off as", across persons, given only partial interpersonal comparability. (This case is controversial, and we'll come back to it.)

"Is at least as good a chess player as", at least arguably (if you think Elo rankings don't capture everything).

"Is at least as good as", all things considered. Again, controversial, but Sen would say it is.

2.7 Failures of transitivity, quasi-transitivity and acyclicity

Majority preference on a Condorcet profile. Three voters ranking $x > y > z$, $y > z > x$, $z > x > y$. The social R is complete and reflexive, but xPy, yPz, zPx: transitivity, quasi-transitivity and acyclicity all fail, and there is no choice set.

"Beat head-to-head" in a sports season. Complete over the pairs that have played; not transitive and typically not even acyclic. Every league produces cycles, which is why leagues rank on points rather than on the beating relation.

Just-noticeable-difference cases. "Tastes at least as sweet as" over cups of coffee differing by one grain of sugar: I is not transitive (each adjacent pair is indifferent, the ends are not), but P is. (Again controversial, but intuitively is a case.)

Supermajority rules. A $2/3$ rule yields a P that is not quasi-transitive but is acyclic provided the threshold is high enough relative to the number of alternatives and enough disagreement.

The Pareto-extension rule. Quasi-transitive but not transitive.

2.8 Maximality without bestness

Choosing between two job offers that are incomparable. Neither offer is at least as good as the other, so neither is best; both are maximal. The maximal set has two members, the choice set is empty. This is the practical shape of the $C(S, R) \subseteq M(S, R)$ result.

Pareto-optimality itself. There are typically very many Pareto-optimal states and no Pareto-best one. Lemma 2*e guarantees at least one of the former over a finite set; nothing guarantees any of the latter.

2.9 Properties α and β

α, satisfied: the world chess champion, if Pakistani, is the champion of Pakistan (Sen's own gloss).

α, violated: the much-discussed menu case — choosing the chicken from {chicken, fish} but the fish from {chicken, fish, frog's legs}. Big question is whether this is an α violation or a change of optiosn.

β, violated: we'll come back to examples of this.

Part 3: Results

Proofs omitted throughout.

3.1 Chapter 1*

Lemma 1*a. If R is a quasi-ordering, then for all $x, y, z \in S$:

(1) $xIy \wedge yIz \rightarrow xIz$;
(2) $xPy \wedge yIz \rightarrow xPz$;
(3) $xIy \wedge yPz \rightarrow xPz$;
(4) $xPy \wedge yPz \rightarrow xPz$.

(These are II, PI, IP, PP respectively.)

Lemma 1*b. Any finite quasi-ordered set has at least one maximal element.

Lemma 1*c. If R is reflexive, then $xPy \leftrightarrow [x] = C(\{x, y\}, R)$.

Lemma 1*d. If for a quasi-ordering R, $C(S,R)$ is non-empty, then $C(S,R) = M(S,R)$.

Lemma 1*e. For any quasi-ordering R over a finite set S: $\forall x,y : [x,y \in M(S,R) \rightarrow xIy] \leftrightarrow [C(S,R) = M(S,R)]$.

Lemma 1*f. For every quasi-ordering Q, there is an ordering R compatible with Q. (Szpilrajn 1930.)

Lemma 1*g. If Q is a quasi-ordering such that $\forall x,y \in S \subset X : xQy \leftrightarrow x = y$, and T is an ordering of elements of S, then there is an ordering R of all elements of X compatible with both Q and T. (Arrow 1951.)

Lemma 1*h. If a quasi-ordering Q_1 is a subrelation of a quasi-ordering Q_2, then Q_1 and Q_2 are compatible.

Lemma 1*i. If Q is a quasi-ordering such that $\forall x,y \in S \subset X : xQy \leftrightarrow x = y$, and T is a quasi-ordering of elements of S, then there is an ordering R of all elements of X compatible with both Q and T.

Lemma 1*j. If R is an ordering defined over a finite set X, then a choice function $C(S,R)$ is defined over X.

Lemma 1*k. If R is reflexive, complete and quasi-transitive over a finite set X, then a choice function $C(S,R)$ is defined over X. (Sen 1969, Theorem II; Pattanaik 1968a.)

Lemma 1*l. If R is reflexive and complete, then a necessary and sufficient condition for $C(S,R)$ to be defined over a finite X is that R be acyclical over X.

Lemma 1*m. Every choice function $C(S,R)$ generated by a binary relation R satisfies property α but not necessarily property β.

Lemma 1*n. A choice function $C(S,R)$ generated by a binary relation R satisfies property β if and only if R is PI-transitive. (Sen 1969, Theorem III.)

Lemma 1*o. (a) In general, PP and PI are completely independent of each other. (b) Together, PP and PI imply transitivity, given completeness of R.

Lemma 1*p. If a binary relation R generates a choice function, then PI-transitivity implies that R is an ordering. (Sen 1969, Theorem IV.)

Lemma 1*q. A choice function $C(S,R)$ derived from a binary relation R satisfies property β if and only if R is an ordering.

Lemma 1*r. If R is complete, then

(a) $PI \leftrightarrow IP$;
(b) $PI \rightarrow II$;
(c) $PP \wedge II \rightarrow PI$.

3.2 Chapter 2*

Lemma 2*a. The relation $\bar{R}$ is a quasi-ordering for every logically possible combination of individual preferences.

Lemma 2*b. A necessary and sufficient condition for $\bar{R}$ to be an ordering, and for $R = \bar{R}$ to be a decisive collective choice rule, is that

$$\forall x, y \in X : [(\exists i : xP_i y) \rightarrow (\forall j : xR_j y)]$$

Lemma 2*c. Both $\bar{P}$ and $\bar{\bar{P}}$ are strict partial orderings (transitive and asymmetric) for every logically possible combination of individual preferences.

Lemma 2*d. $\forall x, y \in X : x\bar{\bar{P}}y \rightarrow x\bar{P}y$.

Lemma 2*e. For every set of individual preferences $(R_1, ..., R_n)$ over any finite set of social states S, there is at least one Pareto-optimal state.

Lemma 2*f. The Kaldor test is inconsistent with every possible CCR under some configuration of preferences. (Scitovsky 1941.)

That is: there are profiles on which the Kaldor test delivers both xPy and yPx, so it does not even define a relation, let alone a well-behaved one.

Lemma 2*g. The Scitovsky compensation test may yield an intransitive P. (Gorman 1955.) It may fail even quasi-transitivity.

Lemma 2*h. If for all $x, y \in X$:

$$[\exists x' \in S(x) : x'\bar{P}y] \rightarrow \forall y' \in S(y) : [\exists x'' \in S(x) : x''\bar{R}y']$$

then P, under the Scitovsky test, is a strict partial ordering. (Cf. Samuelson 1950b: the condition is a complete outward movement of the utility possibility frontier.)

Under that same condition the Kaldor and Scitovsky preference relations coincide, since $xPy \rightarrow \sim (yPx)$ already holds for Kaldor.

3.3 Chapter A1*

T.A1*.1 (Arrow's Impossibility Theorem). No social welfare function can simultaneously satisfy U, I^2, D and P.

T.A1*.2 (Spread of Decisiveness). If G is decisive over any pair $\{x, y\}$, then G is globally decisive.

T.A1*.3 (Contraction of Decisive Sets). If a set G of individuals is decisive, and has more than one member, then a proper subset of G is decisive as well.

Arrow's theorem then follows immediately: by P the set of all individuals is decisive; by T.A1*.3 some proper subset is; iterate; finiteness delivers a single individual, violating D.

T.A1*.4 (Quasi-transitive Oligarchy Theorem). If H is finite and $\#X \geq 3$, then any SDF generating quasi-transitive social preference relations, satisfying U, P and I^2, must be oligarchic. (Gibbard 1969.)

Veto-Spread lemma. Any person who is semi-decisive over one ordered pair must be semi-decisive over all ordered pairs — i.e. must have a veto.

Together with T.A1*.2, which continues to hold under quasi-transitivity, this gives T.A1*.4: no group other than a superset of the oligarchy G can be decisive, since every member of G has a veto.

Mas-Colell and Sonnenschein (1972). A veto result with acyclicity, without demanding quasi-transitivity. Extended by Blair et al. (1976) to triple acyclicity (no cycles over triples rather than over any subset).

T.A1*.5 (Acyclic neutral monotonicity vetoer theorem). If $\#X \geq \#H$, with H finite, then any SDF satisfying U and NIM must yield someone with a veto. (Blau and Deb 1977.)

T.A1*.6 (Arrow Impossibility for semi-transitivity). If H is finite and $\#X \geq 4$, then there is no SDF satisfying U, I, P and D and yielding semi-transitive social preference.

T.A1*.7 (Arrow impossibility for interval order property). If H is finite and $\#X \geq 4$, then there is no SDF satisfying U, I, P and D and yielding interval-ordered social preference.

Since a semi-order is both semi-transitive and an interval order, the impossibility holds a fortiori for semi-orders. Generalising to s-and-t orders: the result goes through provided $\#X \geq s + t + 2$. Orderings are $s + t = 1$ and need $\#X \geq 3$; semi-orders are $s + t = 2$ and need $\#X \geq 4$. For an infinite X the SDF may be confined to the doubly infinite *union* of sets of s-and-t orders.

The filter results. For $f(\cdot)$ satisfying U, P and I:

1. acyclicity implies that Ω is a pre-filter;
2. quasi-transitivity implies that Ω is a filter;
3. semi-order properties imply that Ω is an ultra-filter;
4. transitivity implies that Ω is an ultra-filter.